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MANUFACTURING INDUSTRIES

Geography - Class 10 | Chapter 6 | Deep
Structured NCERT Notes

SECTION 1: INTRODUCTION TO MANUFACTURING

1.1 What is Manufacturing? - Complete Definition and Understanding

Manufacturing is the process of producing goods in large quantities after processing raw materials into finished products. It involves **converting raw materials that have low value into finished products that have high value.** Manufacturing is considered the **backbone of economic development** because it transforms agriculture and mineral resources into useful products for human consumption and industrial use.

Manufacturing is classified as a secondary activity because it **processes raw materials obtained from primary activities** like agriculture, mining, and forestry. Agriculture provides **raw materials like cotton, sugarcane, jute, and wool** for textile and food processing industries. Mining provides **mineral ores like iron ore, bauxite, and limestone** for steel, aluminum, and cement industries.

Manufacturing is essential for economic development because it **reduces dependence on agriculture, increases national income, modernizes agriculture, and creates multiplicity of jobs** in factories, transportation, trade, services, and infrastructure development. Countries with **strong manufacturing sectors** usually have **higher growth, better living standards, and lower poverty**.

1.2 Importance of Manufacturing Industries - Comprehensive Analysis

Manufacturing industries contribute approximately 17% to India's Gross Domestic Product (GDP).

Manufacturing is a major sector of the economy because **manufacturing productivity per worker is much higher** than agriculture. The national goal is to increase manufacturing growth so that it can accelerate economic development.

Manufacturing drives economic growth by attracting **domestic investment and foreign direct investment**, generating **tax revenue**, and creating **linkages with other sectors**. For example, the automobile industry creates demand for **steel, rubber, glass, electronics, and services**.

Manufacturing provides employment to millions of people across India. It creates **direct employment** in factories and **indirect employment** in transport, trade, maintenance, and services. It provides jobs for **skilled and unskilled workers** and offers more stable income than agriculture.

Manufacturing reduces dependence on agriculture by creating **non-agricultural employment opportunities**. Since a large share of India's population depends on agriculture, manufacturing helps reduce pressure on land and allows rural people to earn income through industrial activities.

Manufacturing modernizes agriculture by producing **tractors, harvesters, threshers, irrigation pumps, fertilizers, pesticides, and storage equipment**. This creates a strong link between agriculture and industry and increases agricultural productivity.

Manufacturing brings technology and innovation into the economy through **modern machinery**,

production techniques, quality control, and research and development. It also improves living standards by producing **consumer goods** like clothes, vehicles, appliances, phones, furniture, and medicines.

Manufacturing produces goods for export, earning **foreign exchange** for the country. Major manufacturing exports include **textiles, garments, jewelry, engineering goods, chemicals, pharmaceuticals, and automobile components.**

Manufacturing supports regional development by setting up industries in backward regions. Industries bring **investment, employment, roads, electricity, water supply, and communication facilities** to surrounding areas.

1.3 Classification of Industries - Complete Framework

Industries are classified on the basis of source of raw materials, primary role, capital investment, and ownership. This classification helps students understand the function, location, scale, and control of different industries.

Agro-based industries use plant and animal-based products as raw materials. Examples include **food processing, cotton textile, jute textile, silk textile,**

woolen textile, leather, paper, tea processing, coffee processing, and rubber processing. These industries are usually located near agricultural regions because raw materials may be **bulky or perishable**.

Mineral-based industries use mineral ores as raw materials. Examples include **iron and steel industry, cement industry, aluminum smelting, machine tools, petrochemicals, and non-ferrous metal industries.** These industries are generally **capital-intensive** and often located near mineral deposits.

Basic industries supply raw materials to other industries. They are also called **key industries or parent industries.** Examples include **iron and steel, copper smelting, aluminum smelting, and petrochemicals.**

Consumer industries produce goods for direct consumption. Examples include **sugar, textiles, food processing, paper, soap, toothpaste, cosmetics, and electronics.** These industries are often located near **consumer markets.**

Small-scale industries have lower investment and provide more employment per unit of investment. Examples include **handlooms, handicrafts, small food processing, leather products, paper products, rubber products, and small engineering units.**

Large-scale industries require large capital and advanced technology. Examples include **iron and steel, cement, automobiles, petrochemicals, shipbuilding, aircraft manufacturing, and power generation.**

Public sector industries are owned and operated by the government. Examples include **SAIL, BHEL, NTPC, ONGC, IOC, BEL, HAL, and railway undertakings.** They focus on strategic sectors, employment, regional balance, and public welfare.

Private sector industries are owned and operated by individuals or private companies. Examples include **Tata Steel, Reliance Industries, Mahindra & Mahindra, Tata Motors, Adani Group, Maruti Suzuki, and Asian Paints.** These industries are **profit-oriented** and operate under market competition.

Joint sector industries are jointly run by government and private individuals. They combine **government support** with **private efficiency** and are common in public-private partnership projects.

Co-operative sector industries are owned and operated by producers or workers. Members share profits and make decisions democratically. Examples include **Amul, sugar cooperatives, co-operative textile mills, and handloom cooperatives.**

SECTION 2: AGRO-BASED INDUSTRIES

2.1 Cotton Textile Industry - Complete Analysis

Cotton textile industry is the most important textile industry in India and one of the **oldest industries in the country**. It uses **cotton as raw material** and produces **yarn and fabric** through spinning, weaving, and finishing processes.

Cotton textile industry contributes significantly to India's economy. It contributes to **industrial production, GDP, employment, and export earnings**. It is one of the largest employment-generating industries after agriculture.

Cotton textile industry in India has ancient origins because India was famous for cotton textiles from ancient times. The **first successful cotton textile mill was established in Mumbai in 1854**. After independence, the industry grew rapidly but faced competition from **powerlooms and synthetic fibres**.

Cotton textile production includes several stages: ginning, cleaning, carding, combing, spinning, weaving, and finishing. Cotton bolls are separated from seeds, spun into yarn, woven into cloth, and then

bleached, dyed, or printed.

Location factors for cotton textile industry include **availability of cotton, humid climate, port facilities, cheap labour, market access, and power supply.**

Humidity is important because cotton fibres break easily in dry climate during spinning.

Mumbai is known as the Cottonpolis of India because it became the largest cotton textile centre. It had advantages of **port facility, humid climate, proximity to cotton-growing areas, labour, capital, and market.**

Ahmedabad is called the Manchester of India because it became an important cotton textile centre. It had advantages of **cotton-growing areas nearby, humid climate, Sabarmati River water supply, labour, and market.**

Other important cotton textile centres include **Surat, Kanpur, Coimbatore, Madurai, Nagpur, Indore, Delhi, and Chennai.** Surat is famous for **synthetic textiles and powerloom production.**

Problems of cotton textile industry include **unreliable power supply, low labour productivity, worn-out machinery, competition from synthetic fibres, small domestic market, and need for modernization.** These problems reduce the global competitiveness of Indian

cotton textiles.

Government support includes **Technology Upgradation Fund Scheme, Integrated Textile Parks, export incentives**, and schemes for textile modernization and cluster development.

2.2 Jute Textile Industry - Complete Analysis

Jute textile industry uses jute as raw material and produces **gunny bags, ropes, mats, carpets, and other jute products**. Jute is called **golden fibre** because of its golden colour and economic importance.

India is one of the leading producers of jute and West Bengal is the most important jute-producing region. Jute is grown mainly in the **Ganga-Brahmaputra delta** because it needs **high temperature, heavy rainfall, humid climate, and alluvial soil**.

Jute fibre is obtained from the jute plant. The plant is harvested, retted in water, stripped, dried, spun into yarn, and woven into jute products. Jute processing requires **abundant water**.

Location factors for jute industry include **proximity to jute-producing areas, water transport, abundant water, cheap labour**, and **Kolkata port**. Jute is bulky and perishable, so mills are located near jute farms.

The Hooghly basin is the main jute textile region. Important centres include **Kolkata, Rishra, Chandernagore, Bhatpara, Naihati, Barrackpore, and Plassey.** The region has raw material, water, labour, river transport, and port facilities.

Problems of jute textile industry include **competition from synthetic substitutes, competition from Bangladesh, small domestic market, outdated machinery, and high production cost.**

Government support includes **National Jute Policy, mandatory jute packaging, export incentives,** and research through jute institutions.

2.3 Sugar Industry - Complete Analysis

Sugar industry processes sugarcane into sugar and molasses. India is one of the leading producers of sugar in the world. Sugar industry is important for **rural economy, employment, transport, and agro-based industrial development.**

Sugar industry is seasonal because sugarcane is harvested in a particular season and must be crushed quickly. Sugarcane is **bulky and perishable**, and it loses sucrose if not processed within **24-48 hours.**

Raw material for sugar industry is sugarcane. Sugarcane requires **hot and humid climate, 75-100 cm**

rainfall, and **fertile soil**. **Black soil and alluvial soil** are suitable for sugarcane cultivation.

Location factors for sugar industry include **proximity to sugarcane fields, cool climate, water availability, transportation facilities**, and **labour availability**. Mills must be close to farms because sugarcane is heavy and perishable.

Uttar Pradesh is a major sugar-producing state with centres such as **Meerut, Moradabad, Bareilly, Saharanpur, Lakhimpur Kheri, and Muzaffarnagar**. It has advantages of large sugarcane area, alluvial soil, water, and market.

Maharashtra is another major sugar-producing state with centres such as **Pune, Nashik, Ahmednagar, Kolhapur, Satara, and Solapur**. Maharashtra has efficient cooperative sugar mills, better management, and a suitable climate.

Problems of sugar industry include **seasonal nature, old and inefficient machinery, low yield of sugarcane, delays in payment to farmers**, and **transportation problems**.

Government support includes **fair price for sugarcane, ethanol blending programme, cooperative sugar mills**, and policy support for modernization.

SECTION 3: MINERAL-BASED INDUSTRIES - DEEP DIVE

3.1 Iron and Steel Industry - Complete Analysis

Iron and steel industry is a basic industry because **all other industries depend on it for raw materials**. Steel is used in **construction, machinery, automobiles, tools, appliances, railways, power plants, and infrastructure**.

Iron and steel industry is called the barometer of industrial development because steel production shows the level of industrialization in a country. A strong steel industry supports growth of engineering, transport, construction, and defence sectors.

Raw materials required for steel production include **iron ore, coking coal, limestone, manganese, water, and power**. Iron ore is the main raw material, coking coal is used for smelting, limestone removes impurities, and manganese makes steel hard and durable.

For producing one ton of steel, large quantities of **iron ore, coking coal, limestone, and manganese** are needed. This explains why steel plants are generally located near **iron ore and coal mines**.

Location factors for iron and steel industry include **proximity to raw materials, transportation facilities, water supply, power supply, labour availability, and market access**. Since raw materials are heavy, transportation cost strongly affects location.

Tata Steel at Jamshedpur is the first major private steel plant in India. It has advantages of **iron ore from Odisha, coal from Jharia, water from Subarnarekha River, labour, and transport facilities**.

Major public sector steel plants under SAIL include **Bokaro, Rourkela, Durgapur, Bhilai, and Burnpur**. These plants were set up with foreign assistance and became important centres of industrial development.

Bhilai Steel Plant in Chhattisgarh is one of the largest SAIL plants. It has advantages of **iron ore, coal, water, power, and transport**.

Visakhapatnam Steel Plant is the first shore-based steel plant in India. It has advantages of **port facility**, which helps in importing coking coal and exporting steel.

Problems of iron and steel industry include **high cost and limited availability of coking coal, low labour productivity, irregular energy supply, inadequate infrastructure, and old technology**.

Government support includes **National Steel Policy**, infrastructure development, export incentives, and modernization support.

3.2 Aluminum Smelting Industry - Complete Analysis

Aluminum smelting industry processes bauxite into aluminum. Aluminum is **light, strong, corrosion-resistant, and a good conductor of electricity and heat.** It is used in **aircraft, automobiles, packaging, construction, electrical wires, and consumer products.**

Bauxite is the main raw material for aluminum industry. Bauxite is processed into **alumina** and alumina is smelted into **aluminum** through electrolysis. The process is **highly energy-intensive.**

Location factors for aluminum industry include **proximity to bauxite deposits, cheap and reliable electricity, transportation facilities, and water supply.** Electricity is the most important factor because aluminum smelting consumes huge power.

Odisha is an important aluminum-producing state because it has large bauxite reserves and power resources. Important companies include **NALCO** and **Vedanta Aluminum.**

Other important aluminum centres include **HINDALCO at Renukoot, BALCO at Korba**, and aluminum plants in regions with bauxite and power availability.

Major problems of aluminum industry include **high electricity cost, environmental impact, large capital requirement**, and dependence on efficient transport and power infrastructure.

3.3 Chemical Industries - Complete Analysis

Chemical industries produce both organic and inorganic chemicals. Chemicals are used in **agriculture, manufacturing, pharmaceuticals, plastics, synthetic fibres, paints, soaps, detergents, batteries, and daily-use products.**

Inorganic chemicals include **sulfuric acid, nitric acid, alkalis, soda ash, and fertilizers.** **Organic chemicals** include **petrochemicals, plastics, synthetic rubber, synthetic fibres, solvents, and paints.**

Chemical industry is rapidly growing in India and contributes significantly to GDP, employment, and exports. It supports many other industries including **agriculture, textiles, automobiles, medicines, and construction.**

Location factors for chemical industry include **raw materials, power supply, water availability,**

transport facilities, market access, and environmental regulations. Petrochemical industries are often located near **oil refineries and ports.**

Mumbai is a major chemical centre because of **oil refinery, port facilities, power, water, market, and industrial base.** Important products include **petrochemicals, plastics, synthetic fibres, and pharmaceuticals.**

Other major chemical centres include **Pune, Ahmedabad, Vadodara, Jamnagar, Chennai, Kolkata, Delhi region, and Hyderabad.** Gujarat and Maharashtra are especially important for petrochemical and chemical industries.

Problems of chemical industry include **pollution, toxic waste, high energy use, safety risks, and need for strict environmental control.**

SECTION 4: SUMMARY AND KEY NCERT POINTS

Complete Chapter Summary

1. Manufacturing is the backbone of economic development because it converts raw materials into finished products, adds value, contributes to GDP, creates employment, and supports exports.

2. Industries are classified by raw materials, role, capital investment, and ownership. Important categories include **agro-based, mineral-based, basic, consumer, small-scale, large-scale, public sector, private sector, joint sector, and co-operative sector industries.**

3. Major agro-based industries include **cotton textile, jute textile,** and **sugar industry.** They depend on agricultural raw materials and provide employment to large numbers of people.

4. Cotton textile industry is important for industrial production, GDP, employment, and exports. Important centres include **Mumbai, Ahmedabad, Surat, Coimbatore,** and **Kanpur.**

5. Jute textile industry is concentrated in the **Hooghly basin of West Bengal** because of jute-growing areas, water, labour, river transport, and Kolkata port.

6. Sugar industry is located near sugarcane-growing regions because sugarcane is bulky and perishable. Important states are **Uttar Pradesh, Maharashtra, Karnataka, Tamil Nadu,** and **Bihar.**

7. Major mineral-based industries include **iron and steel, aluminum smelting, chemical industry,** and **cement industry.** These industries are capital-intensive and require raw materials, power, water, transport, and

markets.

8. Iron and steel industry is a basic industry and is called the **barometer of industrial development**.

Important plants include **Tata Steel Jamshedpur, Bokaro, Rourkela, Durgapur, Bhilai, Burnpur, and Visakhapatnam**.

9. Industrial location factors include **raw materials, power, water, labour, market, transportation, capital, and government policies**.

10. Industrial pollution includes **air pollution, water pollution, noise pollution, and land degradation**.

Pollution can be controlled through **treatment plants, recycling, filters, silencers, environmental laws, and cleaner technology**.